

The F -Layer in GR Placement

Blind Fisher–Energy Sector Selection for Levi-Civita Transport

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Abstract

Resolution Geometry requires a positive-definite observation geometry, whereas general relativity supplies a Lorentzian metric and its Levi-Civita connection. The admissible placement is therefore not to identify the GR metric with the RG observation metric, but to let Levi-Civita transport act on a finite-resolution observation projector. Earlier placement statements require an observation model F as input. This paper tightens that layer: given an observation channel B , covariance Σ , and a metric-dependent scalar/action Hessian H_g , the resolved sector is constructed by the generalized Fisher–energy eigenproblem

$$Fv_i = \lambda_i H_g v_i, \quad F = B^T \Sigma^{-1} B.$$

The resulting projectors P_R, P_N are then acted on by Levi-Civita transport through the off-diagonal connection component

$$A_{RN} = P_N(\nabla^{\text{LC}} P_R) P_R.$$

A final integration battery checks the complete finite-dimensional chain: metric/action $\rightarrow H_g$, observation $\rightarrow F$, blind sector selection $\rightarrow P_R, P_N$, and a declared transport perturbation $\rightarrow A_{RN}$, split leakage, and Fisher health before/after. The result is a placement and automation statement, not a derivation of GR and not a reconstruction of data absent from the observation channel.

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1 Purpose

The goal is to close the practical gap in the GR placement bridge. The layer separation is:

$$(M, g_{\text{GR}}, \nabla^{\text{LC}}) \quad \text{supplies transport, not the positive-definite RG metric,}$$

while the finite observer supplies an observation channel

$$B : \phi \mapsto y, \quad \Sigma \succ 0.$$

The new construction is

$$\boxed{(M, g, \nabla^{\text{LC}}) + (B, \Sigma, S[\phi; g]) \longrightarrow (F, H_g) \longrightarrow (P_R, P_N) \longrightarrow A_{RN} = P_N(\nabla^{\text{LC}} P_R) P_R.}$$

The paper therefore removes the impression that P_R is a manually chosen split. The experimenter still declares what is observed, but the resolved/null sector is generated from the observation and action geometry.

Non-claim 1.1 (No replacement of GR geometry). The Lorentzian metric g_{GR} is not inserted into the positive-definite RG observation metric slot. GR remains a spacetime theory with its own metric and Levi-Civita connection.

Non-claim 1.2 (No automatic experiment selection). GR alone does not specify the detector, observation map, or covariance. The observation channel B and covariance Σ must be supplied by the measurement model.

Non-claim 1.3 (No recovery from Fisher holes). If F has null or near-null directions, the construction identifies them as unresolved. It does not reconstruct information that the observation channel does not contain.

2 Observation/action geometry

Let a metric-dependent field model induce a finite action

$$S[\phi; g].$$

For the local quadratic geometry relevant to sector selection, define

$$H_g = \frac{\partial^2 S}{\partial \phi^2}.$$

Let B be the observation map and Σ the positive-definite noise/covariance model. The Fisher observation form is

$$\boxed{F = B^T \Sigma^{-1} B.}$$

The pair (F, H_g) determines which field directions are both observable and dynamically meaningful.

Definition 2.1 (Blind Fisher–energy sector). Assume $H_g \succ 0$ and $F \succeq 0$. Solve

$$F v_i = \lambda_i H_g v_i, \quad \lambda_1 \geq \lambda_2 \geq \cdots.$$

For a dimension budget k , set

$$R_k = \text{span}\{v_1, \dots, v_k\}, \quad N_k = R_k^{\perp_{H_g}}.$$

Let P_R, P_N be the corresponding H_g -orthogonal projectors.

Theorem 2.2 (Optimal Fisher capture). *Among all k -dimensional subspaces with H_g -orthonormal basis Q , the top generalized eigenvectors maximize*

$$\text{tr}(Q^T F Q), \quad Q^T H_g Q = I_k.$$

Proof. Let $H_g = LL^T$ and set $U = L^T Q$. Then $Q^T H_g Q = I$ is equivalent to $U^T U = I$, and

$$\text{tr}(Q^T F Q) = \text{tr}(U^T L^{-1} F L^{-T} U).$$

By Rayleigh–Ritz, this trace is maximized by the top eigenvectors of the symmetric matrix $L^{-1} F L^{-T}$. Transforming back gives the generalized eigenvectors of $Fv = \lambda H_g v$. $\square$

Corollary 2.3 (Automation of the projector). *Given (B, Σ, H_g) and a dimension or threshold rule, the resolved projector P_R is determined by the Fisher–energy geometry. The split is therefore an output of the observation/action geometry, not a hand-picked coordinate split.*

3 GR transport coupling

Once P_R is available, GR enters through Levi-Civita transport. Along a path or infinitesimal displacement, the resolved projector changes. The resolved–null transport coupling is

$$A_{RN} = P_N(\nabla^{\text{LC}} P_R) P_R.$$

This is an off-diagonal connection component. It is not by itself a groupoid arrow; its compatible path-ordered exponential gives transport between resolution geometries.

Proposition 3.1 (Layered GR placement). *Given $(M, g, \nabla^{\text{LC}}; B, \Sigma, S[\phi; g])$ with $H_g \succ 0$, the finite-resolution GR placement chain is*

$$(M, g, \nabla^{\text{LC}}; B, \Sigma, S) \longrightarrow (F, H_g) \longrightarrow (P_R, P_N) \longrightarrow A_{RN} = P_N(\nabla^{\text{LC}} P_R) P_R.$$

The Lorentzian metric remains in the GR layer; the RG observation split is constructed in positive-definite Fisher–energy geometry.

4 Fisher health and transport diagnostics

Let λ_i be the Fisher–energy eigenvalues. For threshold τ , define

$$r_\tau = \#\{i : \lambda_i / \lambda_1 > \tau\}, \quad h_\tau = n - r_\tau,$$

with effective rank

$$r_{\text{eff}} = \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2}.$$

The final integration battery reports:

1. $\|A_{RN}\|_F$ and $\|A_{RN}\|_2$;
2. split leakage after finite transport;
3. principal angle between initial and transported resolved sectors;
4. Fisher health before and after transport.

5 Final integration battery

The battery uses a 10×10 finite scalar/action grid, block-average observation map B , $F = B^T B$, and $k = 25$ resolved directions. Three cases are tested.

1. Flat control: transport amplitude $\beta = 0$.
2. FRW-like conformal scalar background: controlled scale-factor transport amplitude $\beta = 0.2350528691$. This is a controlled transport perturbation, not a claim of unique FRW holonomy.
3. Schwarzschild exterior radial/tortoise background: equatorial circular-loop input

$$\beta(r) = 1 - \sqrt{1 - 2M/r}, \quad r = 8M,$$

so $\beta = 0.1339745962$.

5.1 Battery results

Case	β	$\ A_{RN}\ _F$	$\ A_{RN}\ _2$	leakage $\ \cdot\ _F$	sector angle
Flat	0	0	0	0	0
FRW-like	0.2350528691	0.2870522849	0.2350528691	0.2851430350	0.2350528691
Schwarzschild $r = 8M$	0.1339745962	0.1636130379	0.1339745962	0.1632586419	0.1339745962

Finding 5.1 (Transport coupling behaves as expected). The flat control gives zero coupling and zero leakage. The FRW-like controlled perturbation and Schwarzschild holonomy input give nonzero A_{RN} and split leakage proportional to the declared transport amplitude. In the Schwarzschild case, the operator norm of A_{RN} equals the loop holonomy fraction $\beta(8M)$ used to seed the transport block.

5.2 Health before and after

Case	rank before	hole before	eff. rank before	rank after	hole after	eff. rank after
Flat	25	75	19.2168	25	75	19.2168
FRW-like	25	75	18.3764	25	75	18.3764
Schwarzschild	25	75	11.0919	25	75	11.0919

Finding 5.2 (Transport does not hide the Fisher hole). The finite transport used in the battery rotates the resolved sector but preserves the Fisher spectrum in the transported frame. Thus the before/after health vector remains unchanged while split leakage and transport angle record the geometric mixing. The hole is not erased by transport; it remains visible as the unresolved complement.

5.3 Alignment diagnostics

The post-transport algorithm-selected sector aligns with the transported and reselected sector according to the principal-angle diagnostics:

Case	post-transport alignment angle	before/after selected angle
Flat	4.15×10^{-14}	4.15×10^{-14}
FRW-like	0.4701057382	0.2350528691
Schwarzschild	0.2679491924	0.1339745962

The factor-of-two alignment angle appears because the comparison is between two finite transported subspace representatives: the transported initial frame and the freshly reselected Fisher sector after rotating the Fisher form. The physical leakage angle from the initial sector is the declared transport amplitude in this synthetic block model.

6 Interpretation

The final battery connects the previously separate components:

$$g \rightarrow H_g, \quad B, \Sigma \rightarrow F, \quad (F, H_g) \rightarrow P_R, P_N, \quad \nabla^{\text{LC}} P_R \rightarrow A_{RN}.$$

The result is not that GR determines the observation map. The result is that once the measurement model is specified, the sector on which GR transport acts is no longer arbitrary.

7 Conclusion

The F -layer gap in GR placement is closed in the conservative sense. GR supplies Levi-Civita transport. The finite observer supplies an observation channel and covariance. The metric/action supplies H_g . The Fisher-energy generalized eigenproblem supplies P_R, P_N . Levi-Civita transport then supplies $A_{RN} = P_N(\nabla^{\text{LC}} P_R)P_R$. The unresolved sector remains measurable as the Fisher hole and is not falsely treated as resolved information.

A Reproducibility

The integration battery outputs are saved as

- `rg_gr_flayer_integration_battery_results.json`,
- `rg_gr_flayer_integration_battery_summary.txt`.

The battery is deterministic. It uses 10×10 grids, 2×2 block-average observations, mass parameter $m^2 = 0.7$, resolved budget $k = 25$, Schwarzschild loop radius $r = 8M$, and the finite transport block

$$\Omega = \begin{pmatrix} 0 & -B^T \\ B & 0 \end{pmatrix}, \quad B_{ii} = \frac{\beta}{i+1}, \quad i = 0, \dots, 5.$$

The finite transport is $U = \exp(\Omega)$.

References

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- [3] A. N. Tikhonov and V. Y. Arsenin, *Solutions of Ill-posed Problems*, Winston, 1977.